

SQL GROUP BY - Mental Model Notes

1. Think in terms of questions

The purpose of SQL is to answer questions.

Example:

"For each employee, what was the most recent date they received a salary change?"

2. GROUP BY creates logical groups

If you write:

```
GROUP BY emp_no
```

then all salary records of the same employee become one group.

Example:

```
10001 | 60000 | 1998
```

```
10001 | 70000 | 2000
```

```
10001 | 80000 | 2002
```

becomes one group containing all three rows. Aggregate functions operate inside this group.

Example:

```
SELECT emp_no, MAX(from_date) FROM salaries GROUP BY emp_no;
```

Result: 10001 → 2002

3. Don't group by columns you want to aggregate

Wrong:

```
GROUP BY emp_no, from_date
```

Every row becomes its own group, so `MAX(from_date)` simply returns the same date.

Rule: If you group by a column, you usually shouldn't aggregate that same column.

4. Only group by what defines one result row

`GROUP BY emp_no` → One row per employee.

`GROUP BY emp_no, salary` → One row per employee per salary, which often defeats the purpose.

5. Aggregate functions lose row relationships (VERY IMPORTANT)

Example data:

```
50000 | 1999
```

```
70000 | 2000
```

```
65000 | 2002
```

Query:

```
SELECT emp_no, MAX(from_date), MAX(salary) FROM salaries GROUP BY emp_no;
```

Result:

MAX(from_date)=2002
MAX(salary)=70000

But the salary in 2002 was actually 65000. SQL computed the maximum date and maximum salary independently. It did not preserve the relationship between them.

6. Aggregate functions work independently

MAX(from_date) means “find the largest date.”
MAX(salary) means “find the largest salary.”

It does NOT mean “give me the salary from the row having the largest date.”

7. HAVING cannot magically solve this

Filtering with HAVING cannot recover the relationship between the maximum date and its corresponding salary after aggregation.

8. GROUP BY has limitations

GROUP BY is excellent for:

- Maximum salary
- Minimum salary
- Average salary
- Count employees
- Latest date
- Earliest date

It cannot directly answer:
“Give me the salary that belongs to the latest date.”

You'll later learn techniques like Subqueries, CTEs, Window Functions, and Joins.

Mental model to remember ■

GROUP BY first creates groups, aggregate functions summarize each group, and SQL returns one row per group.

Exam / Interview takeaway

1. GROUP BY creates groups.
2. Aggregate functions operate inside each group.
3. Don't group by the column you're trying to aggregate.
4. Aggregate values (MAX, MIN, etc.) don't preserve relationships with other columns.